

Zitian Zhang

PhD Candidate, Université Laval

📍 London, UK & Québec, Canada | ✉ zitian.zhang.1@ulaval.ca

☎ +44-77666-40725 (UK) & +1-418-455-1469 (Canada) | 🌐 [zzt76.github.io](https://github.com/zzt76)

EDUCATION

PhD Candidate in Computer Science, Université Laval, Canada

2023 – Present

- Research with Prof. Jean-François Lalonde, Computer Vision and Systems Lab
- Research interest: Object compositing, image relighting, diffusion models, light representations
- Topic: Understand, represent and recreate lighting via generative models

M.Eng. in Computer Technology, South China University of Technology, China 2020 – 2023

- Research with Assoc. Prof. Chuhua Xian, Multimedia Lab
- Research interest: Consistent depth estimation, indoor light estimation

B.Mgmt. in E-Commerce, Xidian University, China

2016 – 2020

- GPA: 3.7/4.0, top 5%

PUBLICATIONS

- [1] **Z. Zhang**, I. Georgiev, M. Fischer, Y. Hold-Geoffroy, J.-F. Lalonde, and V. Deschaintre, “Unilight: A unified representation for lighting,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, **Highlight**, [\[project page\]](#)
- [2] F. Fortier-Chouinard, **Z. Zhang**, L.-E. Messier, M. Garon, A. Bhattad, and J.-F. Lalonde, “Spotlight: Shadow-guided object relighting via diffusion,” in *International Conference on 3D Vision (3DV)*, 2026, [\[project page\]](#)
- [3] **Z. Zhang**, J. U. Davis, J. P. A. Vu, J. Kuang, and J.-F. Lalonde, “Improving the color accuracy of lighting estimation models,” in *Color and Imaging Conference (CIC)*, 2025, **Oral Presentation**, [\[project page\]](#)
- [4] **Z. Zhang**, F. Fortier-Chouinard, M. Garon, A. Bhattad, and J.-F. Lalonde, “ZeroComp: Zero-shot object compositing from image intrinsics via diffusion,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025, **Oral Presentation**, [\[project page\]](#)
- [5] Y. Liang, **Z. Zhang**, C. Xian, and S. He, “Delving into multi-illumination monocular depth estimation: A new dataset and method,” *IEEE Transactions on Multimedia*, 2024, [\[project page\]](#)
- [6] C. Xian, K. Qian, **Z. Zhang**, and C. C. Wang, “Multi-scale progressive fusion learning for depth map super-resolution,” *arXiv preprint arXiv:2011.11865*, 2020

RESEARCH EXPERIENCE

Lighting Representation Editability, Adobe Research London

Apr. 2026 – Present

- Working on the editability of lighting representations

UniLight (CVPR 2026 Highlight), Adobe Research London

Jun. 2025 – Aug. 2025

- Proposed a unified lighting representation framework that aligns text, images, irradiance, and environment maps into a joint latent space using contrastive learning with an SH prediction task
- Created a multi-modal dataset comprising 72,000+ aligned samples from HDR environment maps, using VLMs for lighting captioning and off-the-shelf methods for intrinsic decomposition and light estimation
- Enabled down-stream applications such as lighting-based retrieval, environment-map generation, and lighting control in diffusion-based image synthesis

SpotLight (3DV 2026), Depix and Université Laval

Oct. 2024 – May 2025

– Used shadow maps as guidance to achieve precise and training-free local relighting

Color in Light Estimation (CIC 33 Oral), Meta and Université Laval

Sep. 2024 – Apr. 2025

– Developed a light estimation pipeline with WB correction models to improve color accuracy

ZeroComp (WACV 2025 Oral), Depix and Université Laval

Sep. 2023 – Aug. 2024

– Tackled the challenge of enabling realistic 3D object compositing without relying on paired composite-scene image datasets

– Designed and implemented a diffusion-based model trained solely on synthetic indoor RGB and intrinsic dataset, while generalizing well across various scenes

– Created an evaluation dataset, featuring automatically generated, realistic object composites

Consistent Depth Estimation, South China University of Technology

Sep. 2020 – Jun. 2023

– Introduced a single-view multi-illumination RGB-D dataset

– Developed a post-processing module, enabling a consistent depth prediction in changing illuminations

PATENTS

Zitian Zhang, Frédéric Fortier-Chouinard, Mathieu Garon, Anand Bhattad, and Jean-François Lalonde. *Systems and Methods for Compositing a Virtual Object in a Background Image*. U.S. Provisional Patent Application N° 63/705,195, filed October 9, 2024. (in application)

SKILLS

Python, C++, PyTorch, Blender, Unreal Engine
Diffusion Models, Image Compositing, Image Light Control, Light Estimation

SERVICE

SIGGRAPH reviewer

Feb. 2026 – Present

3DV reviewer

Aug. 2025 – Present

TVCG reviewer

Nov. 2024 – Present

OTHER WORK EXPERIENCE

Game Developer Intern, Alibaba Lingxi Interactive, China

Jun. 2022 – Aug. 2022

– Independently created a functional and engaging mini simulation game using UE4

– Implemented the scene setup, game logic, UI, and basic behavior trees with blueprint and C++

Rendering Developer Intern, Revobit, China

Dec. 2021 – May. 2022

– Developed high-quality rendering solutions for digital fashion industry

– Optimized the real-time and offline rendering system and customized shader pipelines.